

Tejas R. Jammihal

Bioinformatics Scientist · Computational Biology · Machine Learning · Data Science

Philadelphia, PA · hello@tejasri.io · tejasri.io · [LinkedIn](#) · [GitHub](#) · [Google Scholar](#)

SUMMARY

Bioinformatics scientist with seven years of experience building machine learning methods and production data pipelines for genomic and clinical data. Works end to end, from scoping problems with stakeholders (clinicians and scientists) to shipping validated models and tools they rely on. At Children's Hospital of Philadelphia, builds detection algorithms and deep learning classifiers for genomic variants used in rare pediatric disease diagnosis. Previously studied immunotherapy response in cancer at Dana-Farber and MD Anderson, which led to co-first-author papers in *Nature Cancer* and *Cancer Cell*. Started in pharma R&D at Novartis.

SKILLS

Machine learning & statistics: Deep learning (CNN transfer learning with PyTorch and timm), AutoML (AutoGluon), scikit-learn, random forest, survival analysis (Kaplan–Meier, Cox), regression, clustering, leakage-safe validation and threshold calibration

Genomics: WGS, WES, bulk and long-read RNA-seq, SNV/indel and SV/CNV calling (DRAGEN, GATK, MuTect2, Manta, CNVpytor), copy number and clonality (ABSOLUTE, FACETS), differential expression, GSEA, neoantigen prediction

Programming & workflows: Python (pandas, NumPy, SciPy, Matplotlib), R (tidyverse, Bioconductor, ggplot2), SQL, Bash, Git, Docker, Singularity, WDL, Snakemake, Nextflow

Platforms & data: AWS, GCP/Terra, HPC (SLURM, LSF), Firebase, TCGA, PCAWG, cBioPortal, CCLE, GEO, dbGaP

AI-assisted development: Claude Code for scaffolding, refactoring, testing, and documentation

EXPERIENCE

Bioinformatics Scientist II | Children's Hospital of Philadelphia, Philadelphia, PA

Jun 2023 – Present

- **Rare disease program build-out:** Led short-read bioinformatics for CHOP's Undiagnosed Disease Program from its first case, covering data intake, DRAGEN processing, and variant prioritization. Co-designed an automated data pipeline with the Arcus informatics team that ingests and processes new genomes and loads them into Emedgene for clinical review. Scaled throughput to over 150 genomes, with more than 100 cases reviewed in its first 10 months.
- **Balanced translocation detection:** Developed a high-specificity method to find balanced translocations in short-read WGS, a class of events that standard tools miss. Scaled it to more than 16,000 genomes across five cohorts (1000 Genomes, Birth Defects Biorepository, GREGoR, UDN, and newborn rapid genome sequencing) and uncovered new diagnoses.
- **ML-based variant triage:** Built a human-in-the-loop image classifier that labels candidate copy-number calls as real or artifact from images of the underlying sequencing data. Fine-tuned ImageNet-pretrained CNNs and benchmarked them against an AutoML baseline on about 14,000 expert-labeled calls, with grouped (sample-level) splits to prevent data leakage and a decision threshold tuned so no confirmed calls were missed on validation.
- **Multi-omic integration:** Integrated multi-modal data (genome and transcriptome) in a re-analysis that increased diagnostic yield in Cornelia de Lange syndrome (*npj Genomic Medicine* 2026, 2nd author). Contributed to WAFI, a multi-omic long-read sequencing workflow for rare disease diagnosis (ACMG 2026 oral presentation, 2nd author).
- **Clinical genomics:** Run end-to-end WGS analysis under SOPs, from QC through DRAGEN calling, Manta and CNVpytor SV/CNV calling, annotation, and filtering. Communicate results to clinical stakeholders (geneticists and genetic counselors) in case reviews, and contributed variant analysis to studies in congenital heart disease (*JACC* 2025) and rare immune and metabolic disorders.
- **Engineering practice:** Use Claude Code to scaffold, test, and document research code. Review teammates' code and analyses for reproducibility.

Associate Data Scientist | MD Anderson Cancer Center, Houston, TX

Jan 2022 – Jun 2023

- **Biomarkers of exceptional response:** Led the computational analysis of metastatic kidney cancer patients with durable response to immune checkpoint inhibitors across two standard-of-care regimens (co-first author, *Nature Cancer* 2025). Engineered features from WES and RNA-seq (clonal neoantigen load, tertiary lymphoid structure signatures), using ABSOLUTE and FACETS for tumor purity and clonality, and showed they were associated with response where the standard biomarker, tumor mutational burden, was not.
- **Transcriptomic pipelines:** Designed and validated pipelines that quantify human endogenous retrovirus expression from short- and long-read RNA-seq. Applied them to in vitro, patient-derived xenograft, and clinical lymphoma data to identify prognostic features, validated with survival modeling (multivariable Cox regression).

- **Long-read methods:** Benchmarked long-read sequencing methods for isoform characterization, structural variant detection, and antigen discovery. Wrote analyses and figures for manuscripts and grants, and mentored new hires on neoantigen prediction workflows.

Bioinformatics Analyst | Dana-Farber Cancer Institute, Boston, MA

Sep 2020 – Jan 2022

- **Cross-trial classification:** Trained a random-forest classifier on labeled transcriptomic data from one phase III trial (IMmotion151) and externally validated it on an independent trial (JAVELIN Renal 101), testing whether subtypes predict benefit from immunotherapy versus targeted therapy (co-first author, *Cancer Cell* 2024).
- **DNA repair biomarker:** Identified a homologous-recombination-deficiency mutational signature enriched in metastatic triple-negative breast cancer patients with poorer survival and primary resistance to checkpoint blockade (*JCO Precision Oncology* 2022).
- **Data at scale:** Analyzed high-dimensional genomic data (WES, RNA-seq) from more than 2,000 patients across clinical-trial cohorts and TCGA/PCAWG. Containerized pipelines with Docker and Singularity and ran them with WDL and Snakemake on GCP and AWS, working closely with oncologists and physician-scientists.

Computational Biology Co-op (Full-Time) | Novartis, Respiratory Disease Area, Cambridge, MA Jun 2019 – Sep 2020

- **Time-course expression analysis:** Ran differential expression (limma, limma-voom) and pathway enrichment (fgsea) analyses on public time-course gene expression data from mouse models exposed to cigarette smoke, characterizing how the biological response differs between acute and chronic exposure.
- **Translational relevance:** Compared these response signatures with COPD patient data to assess how well the models reflect human disease and to inform indication expansion for existing drugs.
- **Stakeholder tools:** Built an interactive dashboard (Spotfire) that let non-technical stakeholders explore results, working with translational and drug discovery scientists.

PROJECTS

- **Training Log** (train.tejasrj.io, [code](#)). Local-first web app that uses an LLM (Claude Haiku) to suggest a daily workout from available time, equipment, and history, falling back to a curated library of about 170 exercises when the model is unavailable. Works offline and syncs across devices through Firebase.

EDUCATION

Master of Biotechnology, The Pennsylvania State University, University Park, PA 2018 – 2020

M.Tech & B.Tech, Biological Engineering, Indian Institute of Technology Madras, Chennai, India 2012 – 2017

SELECTED PUBLICATIONS

7 peer-reviewed articles, 12 conference abstracts (ASCO, ESMO, ACMG, USCAP, CIS), and about 130 citations. *Co-first author. Full list on [Google Scholar](#).

1. **Jammihal, T.***, Saliby, R.M.*, Labaki, C., et al., Shukla, S.A. Immunogenomic determinants of exceptional response to immune checkpoint inhibition in renal cell carcinoma. *Nature Cancer* 6, 372–384 (2025).
2. Saliby, R.M.*, Labaki, C., **Jammihal, T.***, et al., Braun, D.A. Impact of renal cell carcinoma molecular subtypes on immunotherapy and targeted therapy outcomes. *Cancer Cell* 42, 732–735 (2024).
3. Rajagopalan, R., **Jammihal, T.**, Jadhav, T., et al., Krantz, I. Multi-omic re-analysis increases diagnostic yield in individuals with Cornelia de Lange syndrome. *npj Genomic Medicine* (2026).
4. Josowitz, R., Woyciechowski, S., Jadhav, T., **Jammihal, T.**, et al., Spinner, N.B. Placental malperfusion is associated with adverse outcomes in congenital heart disease and with genetic variants in placental developmental pathways. *JACC* 86, 1704–1720 (2025).
5. Barroso-Sousa, R., Forman, J., Collier, K.A., Weber, Z., **Jammihal, T.**, et al., Tolaney, S.M. Multidimensional molecular profiling of metastatic triple-negative breast cancer and immune checkpoint inhibitor benefit. *JCO Precision Oncology* 6, e2100413 (2022).

HONORS & SERVICE

- Best Poster, Renal Cancer category (Kidney Cancer Association), ESMO 2023. Selected oral presentations at the Kidney Cancer Research Summit 2022 and ACMG 2026.
- Invited peer reviewer for *Nature Biotechnology*, *JCO Clinical Cancer Informatics*, *Scientific Reports*, and *Therapeutic Advances in Medical Oncology*.